

Impact Statement Card

The Impact Statement Card translates technical benchmark results from the kaggle competition (0.86972 MRR@10) into real-world outcomes for agricultural extension officers and smallholder farmers in Sub-Saharan Africa.

Key Content :

- **The Problem:** Smallholder farmers lose significant crop yield due to misdiagnosed plant diseases, delayed agronomic advice, or ineffective pest management guidance.
- **The AI Solution:** A hybrid, high-precision retrieval system combining lexical matching (BM25) and dense semantic search (BGE-Large) to surface accurate agricultural extension factsheets in top rank positions.
- **Quantified Impact:**
 - **Precision & Speed:** Surfacing top-1 relevant recommendations increases field officers' query resolution accuracy, reducing field response times.
 - **Yield Preservation:** Eliminates "hard negative" confusion (e.g., confusing nitrogen deficiency with viral infection), reducing crop losses due to improper pesticide or fertilizer application.
- **Ethical Guardrails:** Emphasize that the AI is built for decision support, retrieving vetted extension manuals from organizations like FAO, CGIAR, and Plantwise, ensuring extension workers receive grounded, non-hallucinated advice.